

6.2 Biological processes sequester carbon on land and in the oceans on shorter timescales (fast carbon cycle).	a. Phytoplankton sequester atmospheric carbon during photosynthesis in surface ocean waters; carbonate shells/tests move into the deep ocean water through the carbonate pump and action of the thermohaline circulation.
	b. Terrestrial primary producers sequester carbon during photosynthesis; some of this carbon is returned to the atmosphere during respiration by consumer organisms.
	c. Biological carbon can be stored as dead organic matter in soils, or returned to the atmosphere via biological decomposition over several years.